

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1143

Headwaters Wind Farm Li Llc 100 kV
December 26, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1143. The Interconnection Customer proposes a 117.3 MW solar photovoltaic facility interconnecting at the Headwaters Wind Farm Li Llc 100 kV substation in the Indiana area. The request was received on January 29, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 463 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,868,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

THOUSAND SPRINGS to BLOUNT 100 kV: 218.9% loading (130 MVA) following the outage of the WEBSTERVILLE to POWERLINE ELKHORN 100 kV; DFAX = 5.5%.

For the loss of POWERLINE ELKHORN to WEBSTERVILLE 100 kV, the RANCHO SANTA FE SUBSTATION to SNYDER SUBSTATION 100 kV loads to 217.4% of its 128 MVA rating. Project DFAX: 5.5%.

RANCHO SANTA FE to BLOUNT 100 kV: 217.4% loading (124 MVA) following the outage of the POWERLINE ELKHORN to WEBSTERVILLE 100 kV; DFAX = 5.5%.

DUTTON PUMP to BOURNE 100 kV: 172.6% loading (115 MVA) following the outage of the HANSEL to THOUSAND SPRINGS 100 kV; DFAX = 9.2%.

WEBSTERVILLE to ANCHOR HOCKING SUBSTATION 100 kV: 164.6% loading (119 MVA) following the outage of the ANCHOR HOCKING to SEMASS TAP 100 kV; DFAX = 17.9%.

RED LION to BONNY EAGLE 100 kV: 162.5% loading (141 MVA) following the outage of the EATON to CAPE NORTH 100 kV line; DFAX = 5.9%.

For the loss of EAST METHUEN to HENRY COUNTY 500 kV L, the CAPE N to ANCHOR HOCKING 100 kV line loads to 161.4% of its 191 MVA rating. Project DFAX: 22.2%.

For the loss of HANSEL to THOUSAND SPRINGS 100 kV, the BOURNE to KLAMATH CO-GEN 100 kV loads to 159.0% of its 133 MVA rating. Project DFAX: 9.2%.

For the loss of E METHUEN to HENRY COUNY 500 kV L, the EATON to CAPE N 100 kV loads to 158.3% of its 200 MVA rating. Project DFAX: 22.2%.

BONNY EAGLE to OAK PLAINS 100 kV: 156.7% loading (137 MVA) following the outage of the CAPE N to EATON 100 kV; DFAX = 5.9%.

For the loss of HANSEL to THOUSAND SPRINGS 100 kV, the KLAMATH CO-GEN to SHERMAN ST 100 kV loads to 155.2% of its 143 MVA rating. Project DFAX: 9.2%.

For the loss of E METHUEN 500 kV/230 kV XFMR, the MONASHEE to W RIVERSIDE 100 kV loads to 154.3% of its 194 MVA rating. Project DFAX: 7.9%.

For the loss of HANSEL to THOUSAND SPRINGS 100 kV Line, the EASTMONT SUBSTATION to SHERAN ST 100 kV loads to 152.0% of its 151 MVA rating. Project DFAX: 9.2%.

For the loss of POWERLINE ELKHORN to WEBSTERVILLE 100 kV, the SNYDER to SEMASS TAP 100 kV L loads to 145.9% of its 162 MVA rating. Project DFAX: 13.0%.

For the loss of EAST METHUEN 500 kV/230 kV Auto, the MONASHEE to ELDEAN 100 kV loads to 145.7% of its 222 MVA rating. Project DFAX: 7.9%.

For the loss of POWERLINE ELKHORN to WEBSTERVILLE 100 kV, the SEMASS TAP to ANCHOR HOCKING 100 kV loads to 143.5% of its 169 MVA rating. Project DFAX: 13.0%.

For the loss of EAST METHUEN 500 kV/230 kV Auto, the ELDEAN to CARBONDALE NW 100 kV L loads to 140.6% of its 244 MVA rating. Project DFAX: 7.9%.

For the loss of E METHUEN 500/230 kV Auto, the CARBONDALE NW to RICHARD LAKE 100 kV loads to 135.2% of its 271 MVA rating. Project DFAX: 7.9%.

MURDOCK to HEADWATERS WIND FARM II LLC 100 kV: 134.9% loading (189 MVA) following the outage of the EATON to CAPE N 100 kV Line; DFAX = 80.4%.

For the loss of E METHUEN 500/230kV Autotransformer, the RICHARD LAKE to SAVONA 100 kV loads to 130.8% of its 298 MVA rating. Project DFAX: 7.9%.

The BUTLER MUNICIPAL POWER PLANT to SAVONA 100 kV is overloaded to 128.3% of its 316 MVA emergency rating for the loss of the E METHUEN 500 kV/230 kV Autotransformer. The project contributes 7.9% DFAX (9.2 MW) to this overload.

HANSEL to BUTLER MUNICIPAL POWER PLANT 100 kV L: 126.4% loading (331 MVA) following the outage of the E METHUEN 500 kV/230 kV Autotransformer; DFAX = 7.9%.

THOUSAND SPRINGS to HANSEL 100 kV L: 125.1% loading (342 MVA) following the outage of the EAST METHUEN 500 kV/230 kV XFMR; DFAX = 7.9%.

The DUTTON PUMP to LEENDECKER SUBSTATION 100 kV is overloaded to 119.8% of its 186 MVA emergency rating for the loss of the LONETREE to SABANERA 500 kV. The project contributes 6.5% DFAX (7.7 MW) to this overload.

The THOUSAND SPRINGS to POWERLINE ELKHORN SUBSTATION 100 kV is overloaded to 119.5% of its 225 MVA emergency rating for the loss of the E METHUEN 500-230 kV Auto. The project contributes 5.9% DFAX (6.9 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
THOUSAND SPRINGS to BLOUNT 100 kV	Rebuild line with higher-capacity conductor	\$2,828,000	24% (\$694,000)
RANCHO SANTA FE SUBSTATION to SNYDER 100 kV	Rebuild line with higher-capacity conductor	\$8,505,000	24% (\$2,088,000)
RANCHO SANTA FE to BLOUNT 100 kV	Rebuild line with higher-capacity conductor	\$4,036,000	24% (\$991,000)
DUTTON PUMP to BOURNE 100 kV	Rebuild line with higher-capacity conductor	\$2,490,000	15% (\$370,000)
WEBSTERVILLE to ANCHOR HOCKING SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$7,767,000	16% (\$1,279,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
RED LION to BONNY EAGLE 100 kV	Rebuild line with higher-capacity conductor	\$4,214,000	13% (\$558,000)
ANCHOR HOCKING to CAPE N 100 kV	Rebuild line with higher-capacity conductor	\$3,414,000	16% (\$558,000)
KLAMATH COGEN to BOURNE 100 kV	Rebuild line with higher-capacity conductor	\$6,511,000	15% (\$968,000)
EATON to CAPE N 100 kV	Rebuild line with higher-capacity conductor	\$7,227,000	16% (\$1,181,000)
OAK PLAINS to BONNY EAGLE 100 kV	Rebuild line with higher-capacity conductor	\$10,548,000	13% (\$1,397,000)
SHERMAN ST to KLAMATH CO-GEN 100 kV	Rebuild line with higher-capacity conductor	\$5,282,000	15% (\$785,000)
MONASHEE to W RIVERSIDE 100 kV	Rebuild line with higher-capacity conductor	\$3,756,000	11% (\$415,000)
EASTMONT SUBSTATION to SHERAN ST 100 kV	Rebuild line with higher-capacity conductor	\$2,810,000	15% (\$418,000)
SEMASS TAP to SNYDER 100 kV	Rebuild line with higher-capacity conductor	\$3,802,000	16% (\$613,000)
MONASHEE to ELDEAN 100 kV	Rebuild line with higher-capacity conductor	\$4,975,000	11% (\$550,000)
SEMASS TAP to ANCHOR HOCKING 100 kV	Rebuild line with higher-capacity conductor	\$10,197,000	16% (\$1,645,000)
ELDEAN to CARBONDALE NW 100 kV L	Rebuild line with higher-capacity conductor	\$4,101,000	11% (\$453,000)
CARBONDALE NW to RICHARD LAKE 100 kV	Rebuild line with higher-capacity conductor	\$3,688,000	11% (\$407,000)
MURDOCK to HEADWATERS WIND FARM II LLC 100 kV	Rebuild line with higher-capacity conductor	\$13,998,000	33% (\$4,669,000)
RICHARD LAKE to SAVONA 100 kV	Rebuild line with higher-capacity conductor	\$9,216,000	11% (\$1,018,000)
BUTLER MUNICIPAL POWER PLANT to SAVONA 100 kV	Rebuild line with higher-capacity conductor	\$2,178,000	11% (\$241,000)
HANSEL to BUTLER MUNICIPAL POWER PLANT SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$6,925,000	11% (\$765,000)
THOUSAND SPRINGS to HANSEL 100 kV L	Rebuild line with higher-capacity conductor	\$8,007,000	11% (\$885,000)
DUTTON PUMP to LEENDECKER SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$13,120,000	24% (\$3,221,000)
THOUSAND SPRINGS to POWERLINE ELKHORN SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$2,904,000	24% (\$699,000)
	Total allocated to project		\$26,868,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.